# Factsheet: Trigonometric identities (radians)

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**Summary**

A list of trigonometric identities with angles measured in radians.

*The main study guide for this factsheet is Guide: Trigonometric identities (radians). If you would like to know more about these, please read the guide.*

*This factsheet measures angles in radians. For the associated factsheet measuring angles in degrees, please go to Factsheet: Trigonometric identities (degrees).*

## Trigonometric identities

### Periodicity and parity

For all angles $\theta$ and for all whole numbers $k \in \mathbb{Z}$:

$$\begin{aligned}
\cos(-\theta) &= \cos(\theta) \\
\sin(-\theta) &= -\sin(\theta) \\
\tan(-\theta) &= -\tan(\theta) \\
\cos(\theta + 2k\pi) &= \cos(\theta) \\
\sin(\theta + 2k\pi) &= \sin(\theta) \\
\tan(\theta + k\pi) &= \tan(\theta)
\end{aligned}$$

### Pythagorean formulas

For all angles $\theta$

$$\begin{aligned}
\cos^2(\theta) + \sin^2(\theta) &= 1 \\
1 + \tan^2(\theta) &= \sec^2(\theta) \\
\cot^2(\theta) + 1 &= \csc^2(\theta)
\end{aligned}$$

### Sum and difference formulas

For all angles $\alpha, \beta$:

$$\cos(\alpha+\beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$$

$$\cos(\alpha-\beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)$$

$$\sin(\alpha+\beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$$

$$\sin(\alpha-\beta) = \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta)$$

$$\tan(\alpha+\beta) = \frac{\tan(\alpha)+\tan(\beta)}{1-\tan(\alpha)\tan(\beta)}$$

$$\tan(\alpha-\beta) = \frac{\tan(\alpha)-\tan(\beta)}{1+\tan(\alpha)\tan(\beta)}$$

**Double angle formulas**

For all angles $\theta$:

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$$

$$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$$

$$\tan(2\theta) = \frac{2\tan(\theta)}{1-\tan^2(\theta)}$$

**Shift formulas**

For all angles $\theta$:

$$\cos\left(\theta+\frac{\pi}{2}\right) = -\sin(\theta)$$

$$\cos\left(\theta-\frac{\pi}{2}\right) = \sin(\theta)$$

$$\sin\left(\theta+\frac{\pi}{2}\right) = \cos(\theta)$$

$$\sin\left(\theta-\frac{\pi}{2}\right) = -\cos(\theta)$$

$$\cos(\theta\pm\pi) = -\cos(\theta)$$

$$\sin(\theta\pm\pi) = -\sin(\theta)$$

$$\sin(\pi-\theta) = \sin(\theta)$$

$$\cos(\pi-\theta) = -\cos(\theta)$$

**Sine and cosine rules**

For a triangle with corners $A, B, C$, angles $\alpha$, $\beta$, $\gamma$ respectively at those corners, and sides

$a, b, c$ opposite their respective corners, the **sine rule** is

$$\frac{\sin(\alpha)}{a} = \frac{\sin(\beta)}{b} = \frac{\sin(\gamma)}{c}$$

and the **cosine rule** is

$$a^2 = b^2 + c^2 - 2bc\cos(\alpha).$$

**Common values of trigonometric functions**

| Angle $\theta$ | $0$ | $\frac{\pi}{6}$ | $\frac{\pi}{4}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ | $\frac{2\pi}{3}$ | $\frac{3\pi}{4}$ | $\frac{5\pi}{6}$ | $\pi$ |
|---|---|---|---|---|---|---|---|---|---|
| $\sin\theta$ | $0$ | $\frac{1}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | $0$ |
| $\cos\theta$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | $0$ | $-\frac{1}{2}$ | $-\frac{\sqrt{2}}{2}$ | $-\frac{\sqrt{3}}{2}$ | $-1$ |
| $\tan\theta$ | $0$ | $\frac{1}{\sqrt{3}}$ | $1$ | $\sqrt{3}$ | un-def. | $-\sqrt{3}$ | $-1$ | $-\frac{1}{\sqrt{3}}$ | $0$ |

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## Version history

v1.0: created in 08/25 by tdhc.

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